Lab 4: Functional Dependencies

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1 People

1.1 Person

$$\text{Person} = \{\text{pid}, \text{firstName}, \text{lastName}, \text{email}, \text{affiliation}, \text{startDate}, \text{endDate}\}$$

Each attribute determines itself, and pid determines the whole table:

$$\text{pid} \rightarrow \text{Person}$$

Assuming that each email belongs to exactly one person, email is also a candidate key:

$$\text{email} \rightarrow \text{Person}$$

1.2 Student

$$\text{Student} = \{\text{pid}, \text{program}\}$$

Each attribute determines itself, and pid determines the whole table:

$$\text{pid} \rightarrow \text{Student}$$

Student is a specialization (ISA) of Person, so pid is also a foreign key. The inherited attributes (name, email, ...) are not repeated here.

1.3 Employee

$$\text{Employee} = \{\text{pid}\}$$

$$\text{pid} \rightarrow \text{Employee}$$

Employee is a subclass of Person, and pid is its key and a foreign key to Person. There is no non-trivial functional dependency inside this relation.

1.4 Academic, NonAcademic

$$\text{Academic} = \{\text{pid}\}, \qquad \text{NonAcademic} = \{\text{pid}\}$$

$$\text{pid} \rightarrow \text{Academic}$$

$$\text{pid} \rightarrow \text{NonAcademic}$$

Both are subsets of Employee, with pid as key and as foreign key to Employee. There is no non-trivial functional dependency inside these relations.

1.5 Faculty, Administrative, Technical

$$\text{Faculty} = \{\text{pid}\}, \qquad \text{Administrative} = \{\text{pid}\}, \qquad \text{Technical} = \{\text{pid}\}$$

$$\text{pid} \rightarrow \text{Faculty}$$

$$\text{pid} \rightarrow \text{Administrative}$$

$$\text{pid} \rightarrow \text{Technical}$$

Faculty is a subsclass of Academic, while Administrative and Technical are suclasses of NonAcademic. In each relation pid is the key and a foreign key to its parent. There is no non-trivial functional dependency inside these relations.

1.6 Advises

$$\text{Advises} = \{\text{studentPid}, \text{academicPid}\}$$

$$(\text{studentPid, academicPid}) \rightarrow \text{Advises}$$

Each attribute determines itself. (studentPid, academicPid) determines the whole table.

1.7 ReportsTo

$$\text{ReportsTo} = \{\text{subordinatePid}, \text{supervisorPid}\}$$

Each attribute determines itself, and subordinatePid determines the whole table, because each employee reports to at most one other employee:

$$\text{subordinatePid} \rightarrow \text{supervisorPid}$$

The attribute supervisorPid does not determine subordinatePid, since a supervisor may have several subordinates. The dean does not appear as a subordinate(he doesn't report to anyone).

2 Laboratories, Projects, and Budgets

2.1 Laboratory

$$\text{Laboratory} = \{\text{labId}, \text{name}, \text{building}, \text{roomNumber}, \text{discipline}, \text{supervisorPid}\}$$

Each attribute determines itself, and labId determines the whole table:

$$\text{labId} \rightarrow \text{Laboratory}$$

(We could've also added the FD :

$$\text{supervisorPid} \rightarrow \text{Laboratory}$$

if we were granted the fact that each supervisor had EXACTLY ONE laboratory to supervise, which is not the case.)

2.2 Attached

$$\text{Attached} = \{\text{pid}, \text{labId}\}$$

Each attribute determines itself, and (labId, pid) determine the whole table:

$$(\text{labId, pid}) \rightarrow \text{Attached}$$

2.3 ResearchProject

$$\text{ResearchProject} = \{\text{code}, \text{title}, \text{startDate}, \text{endDate}, \text{status}\}$$

Each attribute determines itself, and code determines the whole table:

$$\text{code} \rightarrow \text{ResearchProject}$$

2.4 Participates

$$\text{ResearchProject} = \{\text{pid}, \text{projectcode}, \text{role}\}$$

Each attribute determines itself, and (pid, projectcode) determine the whole table:

$$(\text{pid, projectcode}) \rightarrow \text{Participates}$$

or more specifically :

$$(\text{pid, projectcode}) \rightarrow \text{role}$$

2.5 Budget

Budget = {budgetLine, amountGranted, amountDisbursed, startDate, endDate, managerPid}

Each attribute determines itself, and budgetLine determine the whole table:

$$\text{budgetLine} \rightarrow \text{Budget}$$

(We could've also added the FD :

$$\text{managerPid} \rightarrow \text{Budget}$$

if we were granted the fact that each supervisor had EXACTLY ONE laboratory to supervise, which is not the case.)

2.6 FundsPrj

$$\text{FundsPrj} = \{\text{budgetLine}, \text{projectcode}\}$$

Each attribute determines itself, and (budgetLine, projectcode) determine the whole table:

$$(\text{budgetLine, projectcode}) \rightarrow \text{FundsPrj}$$

2.7 FundsPrj

$$\text{FundsLab} = \{\text{budgetLine}, \text{labId}\}$$

Each attribute determines itself, and (budgetLine, labId) determine the whole table:

$$(\text{budgetLine, labId}) \rightarrow \text{FundsLab}$$

3 Equipment and Certification

3.1 EquipmentModel

EquipmentModel = {modelId, commercialName, manufacturer, category, requiredEnvironment, trainingMa

Each attribute determines itself, and modelId determines the whole table:

$$\text{modelId} \rightarrow \text{EquipmentModel}$$

3.2 EquipmentUnit

EquipmentUnit = {serialNo, modelId, labId, acquisitionDate, purchaseCost, status, portable}

The attributes modelId and labId do not determine any other attribute, except themselves, because a laboratory can contain zero to many equipment units, and an equipment model can correspond to zero to many equipment units.

Each attribute determines itself, and:

$$\text{serialNo} \rightarrow \text{EquipmentUnit}$$

3.3 Certification

$$\text{Certification} = \{\text{code}, \text{title}, \text{issuingAuthority}, \text{validityPeriod}, \text{safetyLevel}\}$$

Each attribute determines itself, and:

$$\text{code} \rightarrow \text{Certification}$$

3.4 Requires

$$\text{Requires} = \{\text{modelId}, \text{certCode}\}$$

$$(\text{modelId}, \text{certCode}) \rightarrow \text{Requires}$$

The attribute modelId does not determine certCode, and certCode does not determine modelId.

Each attribute determines itself.

3.5 Holds

$$\text{Holds} = \{\text{pid}, \text{certCode}, \text{issueDate}, \text{expirationDate}, \text{grade}\}$$

$$(\text{pid}, \text{certCode}) \rightarrow \text{Holds}$$

The attribute pid does not determine certCode, and certCode does not determine pid.

Each attribute determines itself.

4 Reservations

4.1 Reservation

The relation is defined as follows:

$$\text{Reservation} = \{\text{resvId}, \text{submissionTime}, \text{startTime}, \text{endTime}, \text{purpose}, \text{status}, \text{pid}, \text{projectCode}, \text{approverPid}\}$$

Each reservation is identified by a unique reservation identifier. Therefore, resvId determines all the other attributes:

$$\text{resvId} \rightarrow \{\text{submissionTime}, \text{startTime}, \text{endTime}, \text{purpose}, \text{status}, \text{pid}, \text{projectCode}, \text{approverPid}\}$$

The attribute pid does not determine resvId, since a person can make multiple reservations. Similarly, projectCode does not determine resvId, since a research project may be associated with multiple reservations. The attribute approverPid does not determine resvId, since an approver may be assigned to many reservations.

4.2 Reserves

The relation is defined as follows:

$$\text{Reserves} = \{\text{resvId}, \text{serialNo}\}$$

The primary key is the composite key (resvId, serialNo). Therefore, the composite key determines both attributes:

$$(\text{resvId}, \text{serialNo}) \rightarrow \{\text{resvId}, \text{serialNo}\}$$

This is a trivial functional dependency because the attributes on the right-hand side are already present on the left-hand side.

Neither resvId nor serialNo alone determines the other attribute. A reservation may include multiple equipment units, and an equipment unit may be reserved in multiple reservations at different times.

5 Maintenance and Calibration

5.1 Maintenance

The relation is defined as follows:

$$\text{Maintenance} = \{\text{serialNo}, \text{startTS}, \text{endTS}, \text{type}, \text{description}, \text{cost}, \text{outcome}, \text{technicalPid}\}$$

According to MA2, a maintenance operation is identified by the equipment unit serial number and the maintenance start timestamp. Therefore, the composite key determines all the other attributes:

$$(\text{serialNo}, \text{startTS}) \rightarrow \{\text{endTS}, \text{type}, \text{description}, \text{cost}, \text{outcome}, \text{technicalPid}\}$$

Neither serialNo nor startTS alone determines the other attribute. An equipment unit may undergo multiple maintenance operations at different times, and different equipment units may undergo maintenance operations starting at the same timestamp.

Assumption:

We assume that a technical staff member cannot perform maintenance on two different equipment units simultaneously. Therefore, the combination of technicalPid and startTS determines serialNo:

$$(\text{technicalPid}, \text{startTS}) \rightarrow \text{serialNo}$$

Since (serialNo, startTS) determines the other attributes of a maintenance operation, we can also conclude that:

$$(\text{technicalPid}, \text{startTS}) \rightarrow \{\text{serialNo}, \text{endTS}, \text{type}, \text{description}, \text{cost}, \text{outcome}\}$$

This additional functional dependency relies on our assumption, which is not explicitly stated in MA1–MA3.

5.2 Calibration Records

The relation is defined as follows:

$$\text{CalibrationRecord} = \{\text{serialNo}, \text{calibDate}, \text{calibrationType}, \text{result}, \text{nextDueDate}, \text{remarks}\}$$

Each calibration record is uniquely identified by the equipment unit serial number and the calibration date. Therefore, the composite key determines all the other attributes:

$$(\text{serialNo}, \text{calibDate}) \rightarrow \{\text{calibrationType}, \text{result}, \text{nextDueDate}, \text{remarks}\}$$

Neither serialNo nor calibDate alone determines the other attribute. An equipment unit may have multiple calibration records on different dates, and different equipment units may be calibrated on the same date.

6 Consumables, Suppliers, and Stock

6.1 Consumable

The relation is defined as follows:

$$\begin{aligned}\text{Consumable} = \{&\text{consId}, \text{name},\\ &\text{unitOfMeasure}, \text{hazardLevel},\\ &\text{reorderThreshold}\}\end{aligned}$$

Each consumable is identified by a unique identifier. Therefore, consId determines all the other attributes:

$$\begin{aligned}\text{consId} \rightarrow \{&\text{name}, \text{unitOfMeasure},\\ &\text{hazardLevel}, \text{reorderThreshold}\}\end{aligned}$$

6.2 Supplier

The relation is defined as follows:

$$\text{Supplier} = \{\text{suppId}, \text{name}, \text{contactEmail}, \text{phone}\}$$

Each supplier is identified by a unique identifier. Therefore, suppId determines all the other attributes:

$$\text{suppId} \rightarrow \{\text{name}, \text{contactEmail}, \text{phone}\}$$

Assumption:
We assume that each supplier has a unique contact email and a unique phone number. Therefore:

$$\text{contactEmail} \rightarrow \text{suppId}$$

$$\text{phone} \rightarrow \text{suppId}$$

By transitivity, we obtain:

$$\text{contactEmail} \rightarrow \{\text{suppId}, \text{name}, \text{phone}\}$$

$$\text{phone} \rightarrow \{\text{suppId}, \text{name}, \text{contactEmail}\}$$

6.3 Supplies

The relation is defined as follows:

$$\text{Supplies} = \{\text{suppId}, \text{consId}, \text{labId}, \text{unitPrice}\}$$

The primary key is the composite key $(\text{suppId}, \text{consId}, \text{labId})$. Therefore:

$$(\text{suppId}, \text{consId}, \text{labId}) \rightarrow \text{unitPrice}$$

The unit price depends on the exact combination of supplier, consumable, and laboratory.

6.4 Stocks

The relation is defined as follows:

$$\begin{aligned}\text{Stocks} = \{&\text{labId}, \text{consId},\\ &\text{quantityOnHand}, \text{lastRestockDate},\\ &\text{storageCondition}, \text{technicalPid},\\ &\text{monitoringSince}\}\end{aligned}$$

The primary key is the composite key $(\text{labId}, \text{consId})$. Therefore:

$$\begin{aligned}(\text{labId}, \text{consId}) \rightarrow \{&\text{quantityOnHand}, \text{lastRestockDate},\\ &\text{storageCondition}, \text{technicalPid},\\ &\text{monitoringSince}\}\end{aligned}$$

Neither labId nor consId alone determines the other, since a laboratory may stock multiple consumables and a consumable may be stocked in multiple laboratories.

6.5 Consumes

The relation is defined as follows:

$$\text{Consumes} = \{\text{resId}, \text{labId}, \text{consId}, \text{quantityUsed}\}$$

The primary key is the composite key $(\text{resId}, \text{labId}, \text{consId})$. Therefore:

$$(\text{resId}, \text{labId}, \text{consId}) \rightarrow \text{quantityUsed}$$

Neither resId nor $(\text{labId}, \text{consId})$ alone determines quantityUsed, since different reservations may consume different quantities from the same stock.